Customer Purchasing Behavior Analysis with Python

1. Introduction

This report analyzes customer shopping behavior using survey data from Target. The goal is to understand engagement patterns, satisfaction levels, and the effectiveness of recommendations to improve customer experience and business performance.

2. Task 1: Data Cleaning and Preparation

Techniques Used:

- Handling missing values
- Removing duplicates
- Standardizing categorical data
- Data type conversion
- Column renaming and formatting

Key Findings:

- Dataset was cleaned and standardized for accurate analysis
- Inconsistencies in categorical values were corrected
- Data types were properly aligned for analysis

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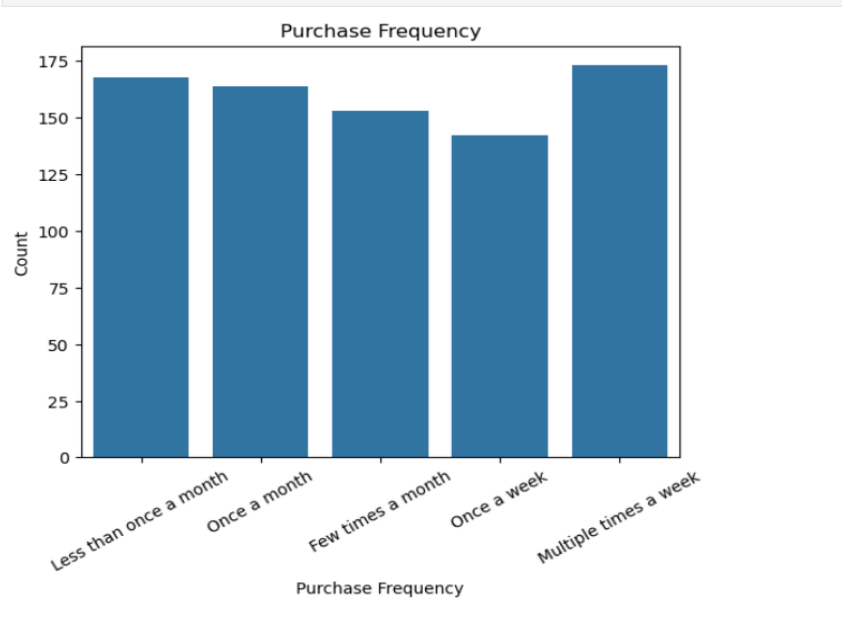
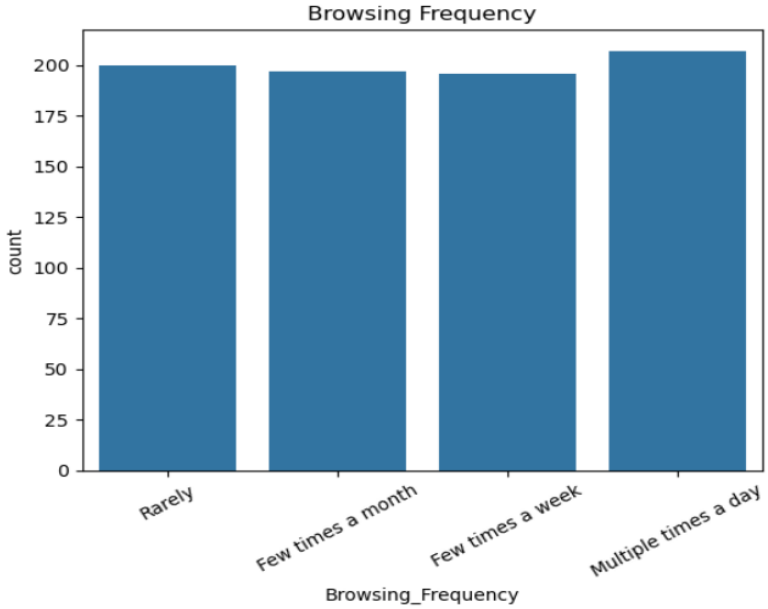
3. Task 2: Descriptive Behavior Analysis

Techniques Used:

- Value counts and frequency distribution
- Statistical measures (mean, median)
- Bar chart visualizations





Key Findings:

- Customer engagement varies across different browsing frequencies
- Certain product categories are more popular
- Cart abandonment is influenced by multiple behavioral factors

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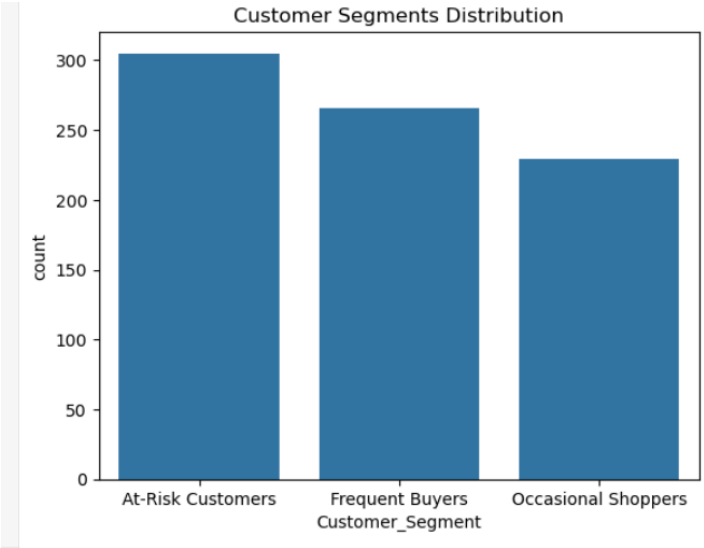
4. Task 3: Customer Segmentation

Techniques Used:

- Grouping and segmentation based on frequency and satisfaction
- (Optional) Clustering techniques like K-Means

Key Findings:

- Identified Frequent Buyers with high satisfaction
- Occasional shoppers show moderate engagement
- At-risk customers have low satisfaction and need attention



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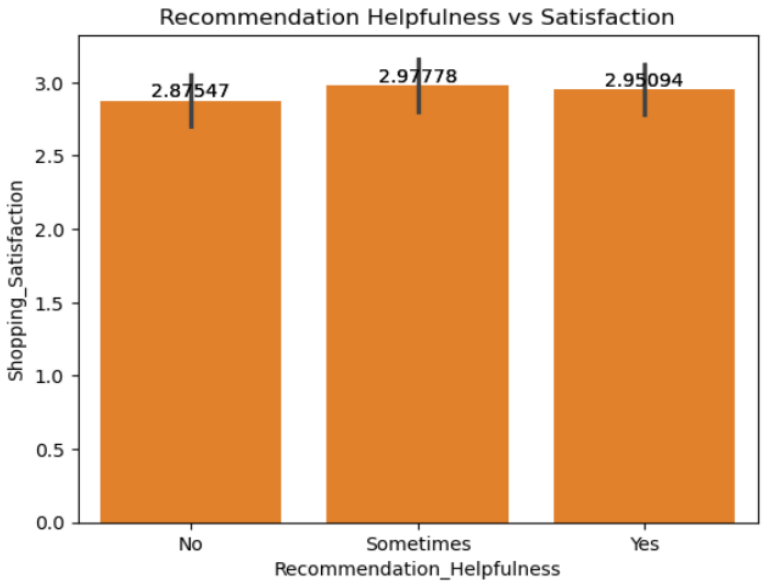
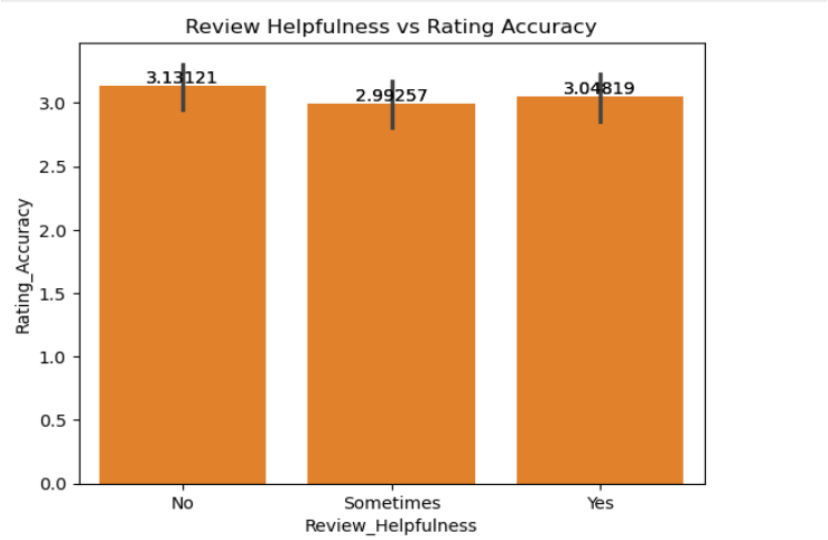
5. Task 4: Recommendation and Review Insights

Techniques Used:

- Comparative analysis
- Relationship analysis between variables





Key Findings:

- Recommendations have limited impact on satisfaction
- Customer trust in reviews varies
- Rating accuracy alone does not drive satisfaction significantly

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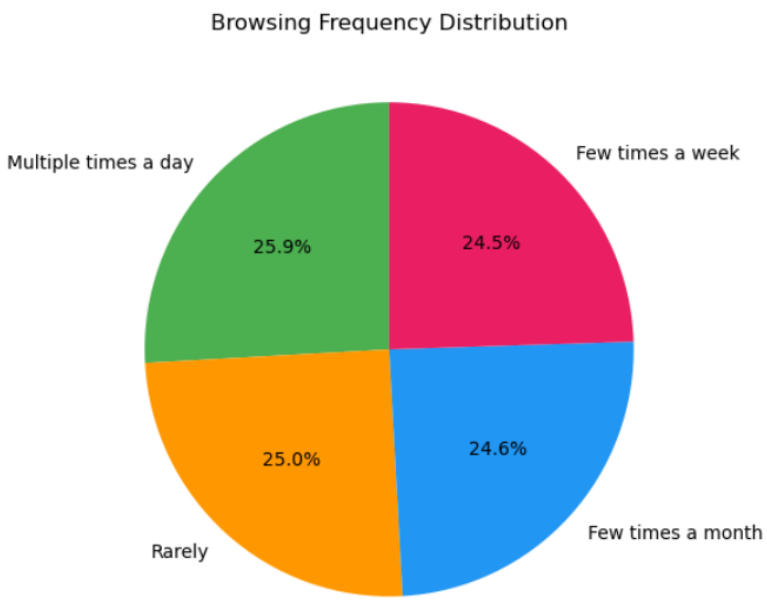
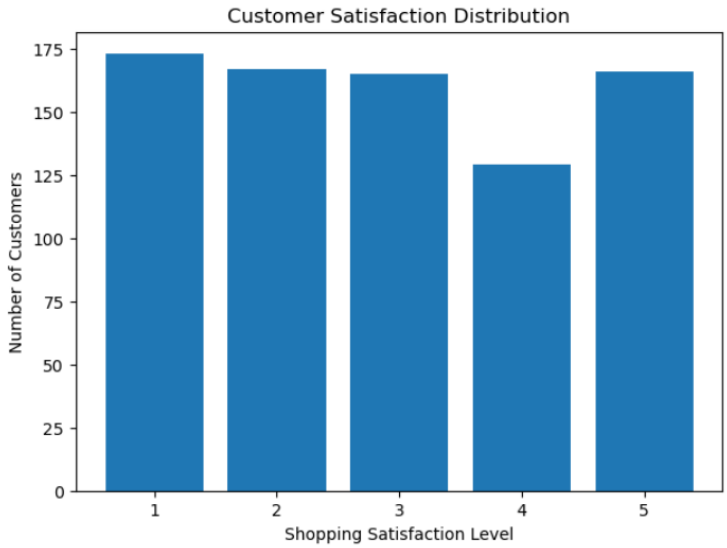
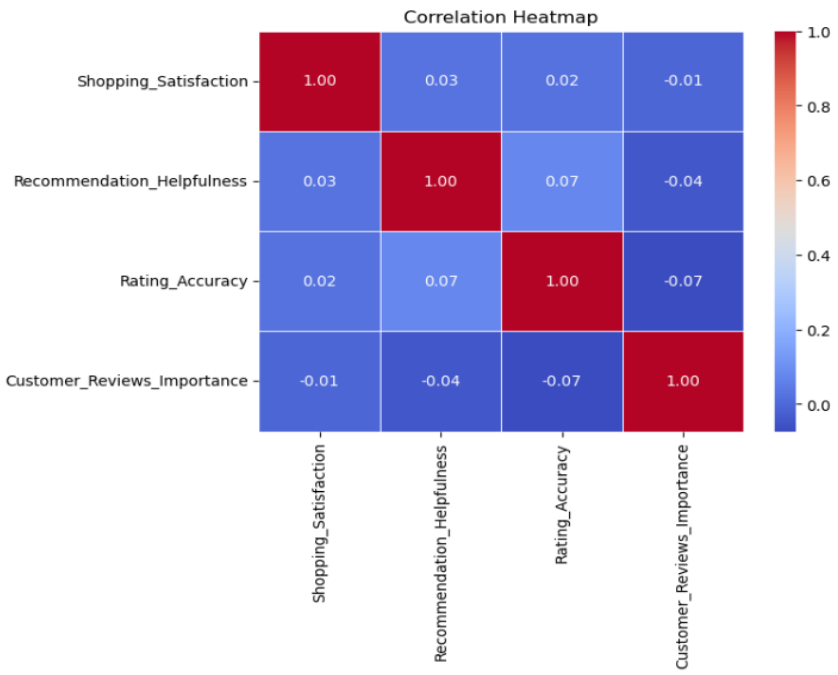
6. Task 5: Visualization and Reporting

Techniques Used:

- Pie charts for distribution
- Bar charts for comparison
- Heatmaps for correlation analysis







Key Findings:

- Browsing frequency is evenly distributed among users
- Satisfaction levels are mixed, with noticeable low ratings
- Correlation between variables is weak, indicating multiple influencing factors

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Overall Key Findings

- Customer engagement is moderate
- Satisfaction is inconsistent across users
- Recommendations and reviews are not major drivers of satisfaction
- Multiple external factors influence customer experience

Key Insights:

-Customer satisfaction is not strongly influenced by recommendations, ratings, or reviews alone.

-Other factors like pricing, delivery, and user experience may play a bigger role.

-There is an opportunity to improve engagement and satisfaction simultaneously.

Recommendations

- Improve personalized recommendation systems
- Enhance user experience and service quality
- Increase engagement using targeted offers
- Strengthen trust in reviews and ratings

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7. Task 6: Video Presentation

Link: https://www.loom.com/share/9272e2b949314a95a536b3b47e617a6f

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Conclusion

The analysis of customer behaviour on the Target platform reveals that while user engagement is moderate, customer satisfaction levels remain inconsistent across the dataset. The browsing patterns indicate a balanced mix of active and inactive users, highlighting opportunities to improve engagement strategies.

The correlation analysis shows that key factors such as recommendation helpfulness, rating accuracy, and customer reviews have very weak influence on overall satisfaction, suggesting that customer experience is driven by multiple factors beyond these variables.

These findings emphasize the need for a holistic improvement strategy, focusing not only on recommendation systems but also on enhancing overall user experience, service quality, and personalization.

By improving engagement for low-frequency users, strengthening trust in reviews, and optimizing platform usability, Target can significantly improve customer satisfaction and retention.